

Vehicle Control System - High Level Design

Prototype Evaluation Document - Synthetic Data

Field	Value
Document ID	HLD-VCS-001
Version	2.0
System	Vehicle Control System
Classification	Synthetic / Evaluation Only

1. Purpose

This High Level Design describes the software architecture of the Vehicle Control System. The document defines software components, ports, interfaces, signals, dependencies and high-level functional flow for prototype evaluation.

2. Software Components

Component	Responsibility
WheelSpeedSensor	Acquires wheel-speed information.
VehicleSpeedController	Processes wheel-speed information and calculates validated vehicle speed.
InstrumentCluster	Displays validated vehicle speed.
DiagnosticsManager	Monitors vehicle-speed plausibility and diagnostic status.

3. Interfaces

Interface	Provider	Consumer
WheelSpeedInterface	WheelSpeedSensor	VehicleSpeedController
VehicleSpeedInterface	VehicleSpeedController	InstrumentCluster
DiagnosticStatusInterface	VehicleSpeedController	DiagnosticsManager

4. Port Definitions

Component	Port	Port Type	Interface
WheelSpeedSensor	WheelSpeedOut	P-Port	WheelSpeedInterface
VehicleSpeedController	WheelSpeedIn	R-Port	WheelSpeedInterface
VehicleSpeedController	VehicleSpeedOut	P-Port	VehicleSpeedInterface
InstrumentCluster	VehicleSpeedIn	R-Port	VehicleSpeedInterface
DiagnosticsManager	DiagnosticStatusIn	R-Port	DiagnosticStatusInterface

5. Signal Definitions

Signal	Interface	Unit
Wheel_Speed	WheelSpeedInterface	km/h
VehicleSpeed	VehicleSpeedInterface	km/h
DiagnosticStatus	DiagnosticStatusInterface	enum

6. Component Dependencies

Dependent Component	Required Component
VehicleSpeedController	WheelSpeedSensor
InstrumentCluster	VehicleSpeedController
DiagnosticsManager	VehicleSpeedController

7. Functional Flow

WheelSpeedSensor -> VehicleSpeedController -> InstrumentCluster -> DiagnosticsManager

8. Revision Information

Current document version: 2.0. Document identifier: HLD-VCS-001.